

TOSHKENT DAVLAT TRANSPORT UNIVERSITETI

TRANSPORTDA AXBOROT TIZIMLARI VA TEXNOLOGIYALARI
Kafedrası

Axborot xavfsizligi va axborotni himoyalash
Fanidan

LABORATORIYA ISHI 14-15

Bajardi: Ibrogimov B.I
Tekshirdi: Sultonova Z.R

Toshkent 2026

Gammalashtirish inversiyalash usuli yordamida

Ishdan maqsad: Ochiq matnlarni simmetrik shifrlashning gammalashtirishning inversiyali usuli algoritmi bilan tanishish va uni amalda qo'llashni o'rganish.

Nazariy qism:

Inversiya usulini amalga oshirish uchun dastavval shifrlanadigan matnning binar obrazi, ya'ni, matn harflarining kompyuter xotirasidagi ikkilik sanoq tizimidagi tasviri yaratiladi.

So'ngra har bir harfning binar obrazining inversiyasi, ya'ni aksini yaratamiz.

Buning uchun xotiradagi har bir ikkilik son o'zining aksi qiymatiga, ya'ni nollar birga, birlar esa nolga almashtiriladi va ulardan kod soni aniqlanib, yangi harf hosil qilinadi.

Quyidagi (6.1-jadval) jadvalda ASCI kodlari keltirilgan bo'lib, biz u asosida har bir son, harf va belgilarning mashina xotirasidagi son kodlari haqida ma'lumot olamiz (o'zbek kirill harflarining ayrimlari bu jadvalga kirmagan).

Shifrlash jarayonini aniq misolda ko'rsak:

Shifrlanuvchi matn:

IBROGIMOV BEHZOD IXTIYORJON UGLI

Kalit so'z: NOK

Gammalashtirish va inversiyalash usulida shifrlash jadvali

Ochiq matn	O'nlik kod ()	Ikkilik ko'rinishi	Inversiyasi (Bitlarni teskarilash)	Kalit (NOK)	Kalit (Ikkilikda)	XOR () natijasi	Shifr kodi (O'nlik)
I	73	01001001	10110110	N (78)	01001110	11111000	248
B	66	01000010	10111101	O (79)	01001111	11110010	242
R	82	01010010	10101101	K (75)	01001011	11100110	230
O	79	01001111	10110000	N (78)	01001110	11111110	254
G	71	01000111	10111000	O (79)	01001111	11110111	247
M	77	01001101	10110010	N (78)	01001110	11111100	252
V	86	01010110	10101001	K (75)	01001011	11100010	226
E	69	01000101	10111010	O (79)	01001111	11110101	245
H	72	01001000	10110111	K (75)	01001011	11111100	252
Z	90	01011010	10100101	N (78)	01001110	11101011	235
D	68	01000100	10111011	K (75)	01001011	11110000	240
X	88	01011000	10100111	O (79)	01001111	11101000	232
T	84	01010100	10101011	K (75)	01001011	11100000	224
Y	89	01011001	10100110	O (79)	01001111	11101001	233
J	74	01001010	10110101	O (79)	01001111	11111010	250
N	78	01001110	10110001	N (78)	01001110	11111111	255
U	85	01010101	10101010	O (79)	01001111	11100101	229
G	71	01000111	10111000	K (75)	01001011	11110011	243
L	76	01001100	10110011	N (78)	01001110	11111101	253

```

14lab.py > ...
1 def gammalashtirish_inversiya_tizimi():
2     # 1. Boshlang'ich ma'lumotlar
3     matn = "IBROGIMOV BEHZOD IXTIYORJON UGLI"
4     tozalangan_matn = matn.replace(" ", "").upper()
5     kalit_soz = "NOK"
6
7     print(f"Ochiq matn: {matn}")
8     print(f"Kalit: {kalit_soz}\n")
9
10    # Kalit kodlarini aniqlash (N=78, O=79, K=75)
11    kalit_kodlari = [ord(k) for k in kalit_soz]
12
13    shifrlangan_kodlar = []
14
15    # --- SHIFRLASH ---
16    print("--- SHIFRLASH JARAYONI ---")
17    print(f"{'Harf':<5} | {'O\'nlik':<6} | {'Ikkilik':<10} | {'Inversiya':<10} | {'K (bin)':<10} | {'XOR (bin)':<10} | {'Shifr':<10}")
18    print("-" * 95)
19
20    for i, harf in enumerate(tozalangan_matn):
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39    # --- DESHIFRLASH ---
40    print("\n--- DESHIFRLASH JARAYONI ---")
41    print(f"{'Shifr':<5} | {'K (bin)':<10} | {'XOR (inv)':<10} | {'Asli (bin)':<10} | {'O\'nlik':<6} | {'Harf':<10}")
42    print("-" * 95)
43
44    tiklangan_matn = ""
45    for i, shifr in enumerate(shifrlangan_kodlar):
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60    print(f"\nNatija: {tiklangan_matn}")
61
62    if __name__ == "__main__":
63        gammalashtirish_inversiya_tizimi()
64

```

1-rasm Dastur kodi

Ochiq matn: IBROGIMOV BEHZOD IXTIYORJON UGLI
Kalit: NOK

--- SHIFRLASH JARAYONI ---						
Harf	O'nlik	Ikkilik	Inversiya	K (bin)	XOR (bin)	Shifr
I	73	01001001	10110110	01001110	11111000	248
B	66	01000010	10111101	01001111	11110010	242
R	82	01010010	10101101	01001011	11100110	230
O	79	01001111	10110000	01001110	11111110	254
G	71	01000111	10111000	01001111	11110111	247
I	73	01001001	10110110	01001011	11111101	253
M	77	01001101	10110010	01001110	11111100	252
O	79	01001111	10110000	01001111	11111111	255
V	86	01010110	10101001	01001011	11100010	226
B	66	01000010	10111101	01001110	11110011	243
E	69	01000101	10111010	01001111	11110101	245
H	72	01001000	10110111	01001011	11111100	252
Z	90	01011010	10100101	01001110	11101011	235
O	79	01001111	10110000	01001111	11111111	255
D	68	01000100	10111011	01001011	11110000	240
I	73	01001001	10110110	01001110	11111000	248
X	88	01011000	10100111	01001111	11101000	232
T	84	01010100	10101011	01001011	11100000	224
I	73	01001001	10110110	01001110	11111000	248
Y	89	01011001	10100110	01001111	11101001	233
O	79	01001111	10110000	01001011	11111011	251
R	82	01010010	10101101	01001110	11100011	227
J	74	01001010	10110101	01001111	11111010	250
O	79	01001111	10110000	01001011	11111011	251
N	78	01001110	10110001	01001110	11111111	255
U	85	01010101	10101010	01001111	11100101	229
G	71	01000111	10111000	01001011	11110011	243
L	76	01001100	10110011	01001110	11111101	253
I	73	01001001	10110110	01001111	11111001	249

2-rasm. Shifrlash natijasi

--- DESHIFRLASH JARAYONI ---					
Shifr	K (bin)	XOR (inv)	Asli (bin)	O'nlik	Harf
248	01001110	10110110	01001001	73	I
242	01001111	10111101	01000010	66	B
230	01001011	10101101	01010010	82	R
254	01001110	10110000	01001111	79	O
247	01001111	10111000	01000111	71	G
253	01001011	10110110	01001001	73	I
252	01001110	10110010	01001101	77	M
255	01001111	10110000	01001111	79	O
226	01001011	10101001	01010110	86	V
243	01001110	10111101	01000010	66	B
245	01001111	10111010	01000101	69	E
252	01001011	10110111	01001000	72	H
235	01001110	10100101	01011010	90	Z
255	01001111	10110000	01001111	79	O
240	01001011	10111011	01000100	68	D
248	01001110	10110110	01001001	73	I
232	01001111	10100111	01011000	88	X
224	01001011	10101011	01010100	84	T
248	01001110	10110110	01001001	73	I
233	01001111	10100110	01011001	89	Y
251	01001011	10110000	01001111	79	O
227	01001110	10101101	01010010	82	R
250	01001111	10110101	01001010	74	J
251	01001011	10110000	01001111	79	O
255	01001110	10110001	01001110	78	N
229	01001111	10101010	01010101	85	U
243	01001011	10111000	01000111	71	G
253	01001110	10110011	01001100	76	L
249	01001111	10110110	01001001	73	I
Natija: IBROGIMOVBEHZODIXTIYORJONUGLI					

3-rasm. Deshifrlash natijasi

RSA algoritmi va elektron raqamli imzo yaratish

Ishdan maqsad: Ochiq matnlarni nosimmetrik shifrlashning RSA usuli algoritmi bilan tanishish va unga elektron raqamli imzo qo'yishni o'rganish.

Nazariy qism:

1977 yilda taklif etilgan RSA algoritmining mualliflari R.Riverst (Rivest), A.Shamir (Shamir) va A.Adleman (Adleman)lar hisoblanishadi. Algoritmning ishonchliligi katta sonlarni faktorlashning (ko'paytuvchilarga ajratish) qiyinligi va diskret algoritmlarni hisoblashning ($a_x = b \pmod{n}$) tenglamada a, b va n aniq bo'lganda x ni topish) qiyinligiga asoslanadi.

RSA algoritmi uchta qismdan iborat: kalitlar generatsiyasi, shifrlash va shifrni o'qish.

IBROGIMOV BEHZOD IXTIYORJON UGLI ochiq matnini RSA shifrlash usuli yordamida shifrlang va unga elektron raqamli imzo qo'ying.

1-bosqich. Kalitlar generatsiyasi

1. **Modulni hisoblash:** $n = p * q = 5 * 7 = 35$
2. **Eyler funksiyasini hisoblash:** $f(n) = (p-1)*(q-1) = 4 * 6 = 24$
3. **Maxfiy kalitni tanlash (d):** d soni $f(n)$ bilan o'zaro tub bo'lishi kerak.
Tanlaymiz: **d = 11**
4. **Ochiq kalitni hisoblash (e):** $(e * d) \bmod f(n) = 1$ shartidan. $(e * 11) \bmod 24 = 1$. Hisoblasak: **e = 11** (chunki $11*11=121$; $121/24$ bo'lsa qoldiq 1 chiqadi).
5. **Xesh boshlang'ich qiymati: H0 = 1**

2-bosqich. Xesh-obrazni hisoblash

Formula: **$H_i = (H_{i-1} + X_i)^2 \bmod n$**

Bunda har bir harfning tartib raqamini (jadval kodi mod 35) olamiz va ketma-ket xesh qiymatini hisoblaymiz. Boshlang'ich qiymat **H0 = 1**.

1. **I** ($73 \bmod 35 = 3$): $H_1 = (1 + 3)^2 \bmod 35 = 16 \bmod 35 = 16$
2. **B** ($66 \bmod 35 = 31$): $H_2 = (16 + 31)^2 \bmod 35 = 2209 \bmod 35 = 4$
3. **R** ($82 \bmod 35 = 12$): $H_3 = (4 + 12)^2 \bmod 35 = 256 \bmod 35 = 11$
4. **O** ($79 \bmod 35 = 9$): $H_4 = (11 + 9)^2 \bmod 35 = 400 \bmod 35 = 15$
5. **G** ($71 \bmod 35 = 1$): $H_5 = (15 + 1)^2 \bmod 35 = 256 \bmod 35 = 11$
6. **I** ($73 \bmod 35 = 3$): $H_6 = (11 + 3)^2 \bmod 35 = 196 \bmod 35 = 21$
7. **M** ($77 \bmod 35 = 7$): $H_7 = (21 + 7)^2 \bmod 35 = 784 \bmod 35 = 14$
8. **O** ($79 \bmod 35 = 9$): $H_8 = (14 + 9)^2 \bmod 35 = 529 \bmod 35 = 4$
9. **V** ($86 \bmod 35 = 16$): $H_9 = (4 + 16)^2 \bmod 35 = 400 \bmod 35 = 15$
10. **B** ($66 \bmod 35 = 31$): $H_{10} = (15 + 31)^2 \bmod 35 = 2116 \bmod 35 = 16$
11. **E** ($69 \bmod 35 = 34$): $H_{11} = (16 + 34)^2 \bmod 35 = 2500 \bmod 35 = 15$
12. **H** ($72 \bmod 35 = 2$): $H_{12} = (15 + 2)^2 \bmod 35 = 289 \bmod 35 = 9$
13. **Z** ($90 \bmod 35 = 20$): $H_{13} = (9 + 20)^2 \bmod 35 = 841 \bmod 35 = 1$
14. **O** ($79 \bmod 35 = 9$): $H_{14} = (1 + 9)^2 \bmod 35 = 100 \bmod 35 = 30$

15. **D** ($68 \bmod 35 = 33$): $H_{15} = (30 + 33)^2 \bmod 35 = 3969 \bmod 35 = \mathbf{14}$
16. **I** ($73 \bmod 35 = 3$): $H_{16} = (14 + 3)^2 \bmod 35 = 289 \bmod 35 = \mathbf{9}$
17. **X** ($88 \bmod 35 = 18$): $H_{17} = (9 + 18)^2 \bmod 35 = 729 \bmod 35 = \mathbf{29}$
18. **T** ($84 \bmod 35 = 14$): $H_{18} = (29 + 14)^2 \bmod 35 = 1849 \bmod 35 = \mathbf{29}$
19. **I** ($73 \bmod 35 = 3$): $H_{19} = (29 + 3)^2 \bmod 35 = 1024 \bmod 35 = \mathbf{9}$
20. **Y** ($89 \bmod 35 = 19$): $H_{20} = (9 + 19)^2 \bmod 35 = 784 \bmod 35 = \mathbf{14}$
21. **O** ($79 \bmod 35 = 9$): $H_{21} = (14 + 9)^2 \bmod 35 = 529 \bmod 35 = \mathbf{4}$
22. **R** ($82 \bmod 35 = 12$): $H_{22} = (4 + 12)^2 \bmod 35 = 256 \bmod 35 = \mathbf{11}$
23. **J** ($74 \bmod 35 = 4$): $H_{23} = (11 + 4)^2 \bmod 35 = 225 \bmod 35 = \mathbf{15}$
24. **O** ($79 \bmod 35 = 9$): $H_{24} = (15 + 9)^2 \bmod 35 = 576 \bmod 35 = \mathbf{16}$
25. **N** ($78 \bmod 35 = 8$): $H_{25} = (16 + 8)^2 \bmod 35 = 576 \bmod 35 = \mathbf{16}$
26. **U** ($85 \bmod 35 = 15$): $H_{26} = (16 + 15)^2 \bmod 35 = 961 \bmod 35 = \mathbf{16}$
27. **G** ($71 \bmod 35 = 1$): $H_{27} = (16 + 1)^2 \bmod 35 = 289 \bmod 35 = \mathbf{9}$
28. **L** ($76 \bmod 35 = 6$): $H_{28} = (9 + 6)^2 \bmod 35 = 225 \bmod 35 = \mathbf{15}$
29. **I** ($73 \bmod 35 = 3$): $H_{29} = (15 + 3)^2 \bmod 35 = 324 \bmod 35 = \mathbf{9}$

Yakuniy xesh-obraz (r) = 9

3-bosqich. RSA usulida shifrlash

Formula: $M_i = (X_i)^e \bmod n$ (Ochiq kalit $e=11$, $n=35$ yordamida)

1. **I** (3): $3^{11} \bmod 35 = \mathbf{12}$
2. **B** (31): $31^{11} \bmod 35 = \mathbf{26}$
3. **R** (12): $12^{11} \bmod 35 = \mathbf{3}$
4. **O** (9): $9^{11} \bmod 35 = \mathbf{9}$
5. **G** (1): $1^{11} \bmod 35 = \mathbf{1}$
6. **I** (3): $3^{11} \bmod 35 = \mathbf{12}$
7. **M** (7): $7^{11} \bmod 35 = \mathbf{7}$
8. **O** (9): $9^{11} \bmod 35 = \mathbf{9}$

9. **V (16):** $16^{11} \bmod 35 = 11$
10. **B (31):** $31^{11} \bmod 35 = 26$
11. **E (34):** $34^{11} \bmod 35 = 34$
12. **H (2):** $2^{11} \bmod 35 = 18$
13. **Z (20):** $20^{11} \bmod 35 = 20$
14. **O (9):** $9^{11} \bmod 35 = 9$
15. **D (33):** $33^{11} \bmod 35 = 12$
16. **I (3):** $3^{11} \bmod 35 = 12$
17. **X (18):** $18^{11} \bmod 35 = 2$
18. **T (14):** $14^{11} \bmod 35 = 14$
19. **I (3):** $3^{11} \bmod 35 = 12$
20. **Y (19):** $19^{11} \bmod 35 = 24$
21. **O (9):** $9^{11} \bmod 35 = 9$
22. **R (12):** $12^{11} \bmod 35 = 3$
23. **J (4):** $4^{11} \bmod 35 = 9$
24. **O (9):** $9^{11} \bmod 35 = 9$
25. **N (8):** $8^{11} \bmod 35 = 22$
26. **U (15):** $15^{11} \bmod 35 = 15$
27. **G (1):** $1^{11} \bmod 35 = 1$
28. **L (6):** $6^{11} \bmod 35 = 6$
29. **I (3):** $3^{11} \bmod 35 = 12$

Shifrlangan kodlar ketma-ketligi:

12, 26, 3, 9, 1, 12, 7, 9, 11, 26, 34, 18, 20, 9, 12, 12, 2, 14, 12, 24, 9, 3, 9, 9, 22, 15, 1, 6, 12.

4.bosqich. Raqamli imzo yaratish

Raqamli imzo yaratish uchun 2-bosqichda hosil bo'lgan yakuniy xesh-obraz (**r**) va maxfiy kalit (**d**) dan foydalaniladi.

Formula: $S = r^d \bmod n$ (Maxfiy kalit $d=11$, $n=35$ yordamida)

Berilganlar:

- Yakuniy xesh-obraz (**r**) = 9
- Maxfiy kalit (**d**) = 11
- Modul (**n**) = 35

Hisoblash:

- $S = 9^{11} \bmod 35$
 $9^1 = 9$
 $9^2 = 81 = 11 \pmod{35}$
 $9^4 = 11^2 = 121 = 16 \pmod{35}$
 $9^8 = 16^2 = 256 = 11 \pmod{35}$
 $S = 9^8 * 9^2 * 9^1 = 11 * 11 * 9 = 121 * 9 = 16 * 9 = 144$
 $S = 144 \bmod 35 = 4$

Hosil bo'lgan elektron raqamli imzo (ERI): $S = 4$

5-bosqich. Deshifrlash va ERI haqiqiyligini tekshirish

1. Matnni deshifrlash (ochish)

- **Formula:** $X_i = M_i^d \bmod n$
- (Maxfiy kalit $d=11$ yordamida har bir kod asliga qaytariladi):
 - M1 (12) uchun: $12^{11} \bmod 35 = 3$ (I harfi)
 - M2 (26) uchun: $26^{11} \bmod 35 = 31$ (B harfi)
 - M3 (3) uchun: $3^{11} \bmod 35 = 12$ (R harfi)
 - M4 (9) uchun: $9^{11} \bmod 35 = 9$ (O harfi)
 - M5 (1) uchun: $1^{11} \bmod 35 = 1$ (G harfi)

2. Raqamli imzoni tekshirish

Qabul qiluvchi olingan imzo (S) va ochiq kalit (e) yordamida tekshiruv xeshini ($r_{\text{tekshiruv}}$) hisoblaydi:

- **Formula:** $r_{\text{tekshiruv}} = S^e \bmod n$

- **Berilganlar:**

- Olingan imzo (S) = 4

- Ochiq kalit (e) = 11

- Modul (n) = 35

- **Hisoblash jarayoni:**

$$r_{\text{tekshiruv}} = 4^{11} \bmod 35$$

$$4^1 = 4$$

$$4^2 = 16$$

$$4^4 = 16^2 = 256 = 11 \pmod{35}$$

$$4^8 = 11^2 = 121 = 16 \pmod{35}$$

$$r_{\text{tekshiruv}} = 4^8 * 4^2 * 4^1 = 16 * 16 * 4 = 256 * 4 = 11 * 4 = 44$$

$$r_{\text{tekshiruv}} = 44 \bmod 35 = 9$$

Tekshiruv natijasi:

- Hisoblangan tekshiruv xeshi: **9**

- Jo'natuvchidan olingan xesh-obraz: **9**

- Natija: **9 = 9**. Elektron raqamli imzo haqiqiy!

15lab.py > ...

```
1 def rsa_eri_tizimi():
2     # 1. Boshlang'ich ma'lumotlar
3     p, q = 5, 7
4     n = p * q
5     phi = (p - 1) * (q - 1)
6     d = 11 # Maxfiy kalit
7     e = 11 # Ochiq kalit ((e * d) % phi == 1)
8     h0 = 1 # Xesh boshlang'ich qiymati
9
10    matn = "IBROGIMOV BEHZOD IXTIYORJON UGLI"
11    tozalangan_matn = matn.replace(" ", "").upper()
12
13    print(f"--- 1-BOSQICH: KALITLAR ---")
14    print(f"p={p}, q={q}, n={n}, phi={phi}")
15    print(f"Ochiq kalit (e): {e}, Maxfiy kalit (d): {d}\n")
16
17    # 2. Xesh-obrazni hisoblash ( $H_i = (H_{i-1} + X_i)^2 \% n$ )
18    print("--- 2-BOSQICH: XESH HISOBLASH ---")
19    h_current = h0
20    for harf in tozalangan_matn:
21        x_i = ord(harf) % n
22        h_next = (h_current + x_i)**2 % n
23        print(f"{harf} (x={x_i}): ({h_current} + {x_i})^2 % {n} = {h_next}")
24        h_current = h_next
25
26    r_xesh = h_current
27    print(f"Yakuniy xesh-obraz (r): {r_xesh}\n")
28
29    # 3. RSA Shifrlash ( $M_i = X_i^e \% n$ )
30    print("--- 3-BOSQICH: RSA SHIFRLASH ---")
31    shifrlangan_kodlar = []
32    for harf in tozalangan_matn:
33        x_i = ord(harf) % n
34        m_i = pow(x_i, e, n)
35        shifrlangan_kodlar.append(m_i)
36    print(f"Shifrlangan matn: {shifrlangan_kodlar}\n")
37
38    # 4. Raqamli imzo yaratish ( $S = r^d \% n$ )
39    print("--- 4-BOSQICH: IMZO YARATISH ---")
40    s_imzo = pow(r_xesh, d, n)
41    print(f"Elektron raqamli imzo (s): {s_imzo}\n")
42
43    # 5. Tekshirish ( $r_{tekshiruv} = S^e \% n$ )
44    print("--- 5-BOSQICH: TEKSHIRISH ---")
45    r_tekshiruv = pow(s_imzo, e, n)
46    print(f"Tekshiruv xeshi (r'): {r_tekshiruv}")
47
48    if r_tekshiruv == r_xesh:
49        print("NATIJA: Imzo haqiqiy, matn butun!")
50    else:
51        print("NATIJA: Imzo xato yoki matn o'zgargan!")
52
53 if __name__ == "__main__":
54     rsa_eri_tizimi()
55
```

4-rasm. Dastur kodi

--- 1-BOSQICH: KALITLAR ---

$p=5$, $q=7$, $n=35$, $\phi=24$

Ochiq kalit (e): 11, Maxfiy kalit (d): 11

--- 2-BOSQICH: XESH HISOBLASH ---

I (x=3): $(1 + 3)^2 \% 35 = 16$
B (x=31): $(16 + 31)^2 \% 35 = 4$
R (x=12): $(4 + 12)^2 \% 35 = 11$
O (x=9): $(11 + 9)^2 \% 35 = 15$
G (x=1): $(15 + 1)^2 \% 35 = 11$
I (x=3): $(11 + 3)^2 \% 35 = 21$
M (x=7): $(21 + 7)^2 \% 35 = 14$
O (x=9): $(14 + 9)^2 \% 35 = 4$
V (x=16): $(4 + 16)^2 \% 35 = 15$
B (x=31): $(15 + 31)^2 \% 35 = 16$
E (x=34): $(16 + 34)^2 \% 35 = 15$
H (x=2): $(15 + 2)^2 \% 35 = 9$
Z (x=20): $(9 + 20)^2 \% 35 = 1$
O (x=9): $(1 + 9)^2 \% 35 = 30$
D (x=33): $(30 + 33)^2 \% 35 = 14$
I (x=3): $(14 + 3)^2 \% 35 = 9$
X (x=18): $(9 + 18)^2 \% 35 = 29$
T (x=14): $(29 + 14)^2 \% 35 = 29$
I (x=3): $(29 + 3)^2 \% 35 = 9$
Y (x=19): $(9 + 19)^2 \% 35 = 14$
O (x=9): $(14 + 9)^2 \% 35 = 4$
R (x=12): $(4 + 12)^2 \% 35 = 11$
J (x=4): $(11 + 4)^2 \% 35 = 15$
O (x=9): $(15 + 9)^2 \% 35 = 16$
N (x=8): $(16 + 8)^2 \% 35 = 16$
U (x=15): $(16 + 15)^2 \% 35 = 16$
G (x=1): $(16 + 1)^2 \% 35 = 9$
L (x=6): $(9 + 6)^2 \% 35 = 15$
I (x=3): $(15 + 3)^2 \% 35 = 9$

Yakuniy xesh-obraz (r): 9

--- 3-BOSQICH: RSA SHIFRLASH ---

Shifrlangan matn: [12, 26, 3, 4, 1, 12, 28, 4, 11, 26, 34, 18, 20, 4, 17, 12, 2, 14, 12, 24, 4, 3, 9, 4, 22, 15, 1, 6, 12]

--- 4-BOSQICH: IMZO YARATISH ---

Elektron raqamli imzo (s): 4

--- 5-BOSQICH: TEKSHIRISH ---

Tekshiruv xeshi (r'): 9

NATIJA: Imzo haqiqiy, matn butun!

5-rasm. Shifrlash natijasi

Xulosa

1. Gammalashtirishning inversiyali usuli:

Mazmuni: Bu usul ochiq matn bitlarini teskarilash (inversiya) va keyin maxfiy kalit (gamma) bilan **XOR** amali orqali birlashtirishga asoslanadi.

Xulosa: Usul simmetrik kriptotizimlarga kiradi. Unda inversiya bitlar tartibini o'zgartirsa, gamma (kalit) shifrnı har bir belgi uchun turlicha ko'rinishga keltiradi. Bu ma'lumotlarni tezkor shifrlash uchun qulay va sodda usuldir.

2. RSA algoritmi va Elektron raqamli imzo (ERI):

Mazmuni: Bu murakkabroq, nosimmetrik usul bo'lib, ochiq va maxfiy kalitlar juftligiga hamda matnning **xesh-obraziga** asoslanadi.

Xulosa: RSA tizimi nafaqat ma'lumotni shifrlaydi, balki uning butunligini va muallifini tasdiqlash imkonini beradi. Xesh-funksiya matnning "barmoq izi"ni yaratsa, ERI ushbu izni maxfiy kalit bilan muhrlaydi. Bu hozirgi kunda bank va davlat xizmatlarida qo'llaniladigan eng ishonchli himoya usullaridan biridir.

Foydalanilgan adabiyotlar

1. G'aniyev S.K., Karimov M.M., Tashev K.A. "Axborot xavfsizligi. Kriptografik usullar". — T.: Aloqachi, 2018.
2. Stallings W. "Cryptography and Network Security: Principles and Practice". — Pearson, 7th Edition, 2017.
3. Schneier B. "Applied Cryptography: Protocols, Algorithms, and Source Code in C". — Wiley, 2nd Edition, 1996.
4. Menezes A., van Oorschot P., Vanstone S. "Handbook of Applied Cryptography". — CRC Press, 1996.
5. Xasanov X.P., Xujakulov A.X. "Axborot xavfsizligi va kriptografiya". — T.: O'zbekiston faylasuflari milliy jamiyati nashriyoti, 2012